Arekanti Siddartha

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Education

SRM University — **Amaravati, Andhra Pradesh**
M.Tech in Cyber Security; ***CGPA: 8.30/10.0*** — *Sep 2025 – Jun 2027 (Expected)*

SRM University — **Amaravati, Andhra Pradesh**
B.Tech in Computer Science and Engineering; ***CGPA: 7.57/10.0*** — *Jun 2021 – May 2025*

Experience

Software Engineering Intern — **May 2024 – Jul 2024**
SECENAI Semiconductors and Test Solutions Pvt. Ltd. — *Amaravati, Andhra Pradesh*

- Architected a Student Attendance System using Face Recognition and OpenCV with 95%+ identification accuracy for 100+ students.
- Automated attendance tracking with Machine Learning algorithms including SVM and CNN, improving efficiency by 80% versus manual methods.
- Engineered real-time facial detection and recognition pipeline using Python, OpenCV, NumPy, and scikit-learn.

Patents & Publications

FEELGAN: Federated Edge Learning with GANs for Medical Image Analysis

- Patent Approved (Application No: 202441083307) – Pioneered a privacy-preserving disease detection system using federated learning, generative adversarial networks, and decentralized healthcare diagnostics.

Projects

FEELGAN - Medical Image Analysis System | *Python, TensorFlow, PyTorch, GANs, CNNs, Federated Learning*

- Developed FEELGAN - Medical Image Analysis System using Python, TensorFlow, PyTorch, GANs, CNNs, Federated Learning, aligned with Python, Tensorflow, Pytorch, CNNs, Image Analysis.
- Implemented Synthetic medical image generation, CNN classification, COVID-19/TB/OCT dataset training, federated learning workflow, improving practical coverage of production-style workflows.
- Delivered measurable outcomes including 15% diagnostic accuracy improvement, 10,000+ synthetic medical images, 90%+ accuracy.

AI-Powered Documentation ChatBot | *Python, React, FastAPI, RAG, LLM, OpenAI, Embeddings*

- Developed AI-Powered Documentation ChatBot using Python, React, FastAPI, RAG, LLM, OpenAI, aligned with Python, Fastapi, REST API, API, Rag.
- Implemented Document parsing, cleaning, chunking, semantic search, improving practical coverage of production-style workflows.
- Delivered measurable outcomes including 1,000+ pages processed, sub-second response times.

Student Attendance System | *Python, OpenCV, NumPy, scikit-learn, SVM, CNN*

- Developed Student Attendance System using Python, OpenCV, NumPy, scikit-learn, SVM, CNN, aligned with Python, Model.
- Implemented Face detection, face recognition, attendance logging, model training, improving practical coverage of production-style workflows.
- Delivered measurable outcomes including 95%+ identification accuracy, 100+ students.

Technical Skills

Languages: Python, C++, HTML/CSS, JavaScript, SQL
Frameworks & Libraries: PyTorch, TensorFlow, Flask, Node.js, OpenCV, React, scikit-learn
Databases: MongoDB, MySQL
Developer Tools: Docker, Git, GitHub, Jupyter Notebook, Postman, PyCharm, VS Code
Core Competencies: CNNs, Computer Vision, Deep Learning, Federated Learning, Machine Learning, REST APIs, GANs

Certifications

Deep Learning with PyTorch – Udemy — **2024**

Machine Learning Specialization – Coursera — **2024**

Full Stack Web Development – freeCodeCamp — **2023**

Achievements

LeetCode – Solved 350+ problems across data structures and algorithms

Anveshan 2024 – Presented FEELGAN at National Student Research Program

Sushacks 2025 – Competed in Nationwide Innovation Hackathon